CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 15: Similarity (With Applications to Maps and Models)

Topic-wise short notes with solved examples

Similar figures have the same shape but different sizes. This chapter covers the tests for similar triangles, the basic proportionality theorem, the ratio of areas, and the use of scale factors in maps and models.

1 Similar triangles

- Two triangles are **similar** if their corresponding angles are equal and their corresponding sides are proportional.
- $\triangle ABC \sim \triangle PQR$ means $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$ and $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$.
- Write the vertices in corresponding order, so that the equal ratios can be read straight from the names.
- Congruent triangles are similar, but similar triangles need not be congruent.

2 Conditions for similarity

Condition	What must be shown
AA (or AAA)	Two angles of one triangle equal two angles of the other
SAS	One angle equal, and the sides including it proportional
SSS	All three pairs of corresponding sides proportional

- In similar triangles, corresponding **altitudes**, **medians** and **perimeters** are in the same ratio as corresponding sides.

Example 2.1 In $\triangle ABC$ and $\triangle DEF$, $\angle A = \angle D$, $AB = 6$ cm, $AC = 9$ cm, $DE = 4$ cm and $DF = 6$ cm. Prove that the triangles are similar, and find EF if $BC = 12$ cm.

SOLUTION

$\frac{AB}{DE} = \frac{6}{4} = \frac{3}{2}$ and $\frac{AC}{DF} = \frac{9}{6} = \frac{3}{2}$, and the included angles A and D are equal.

So $\triangle ABC \sim \triangle DEF$ by SAS.

$\frac{BC}{EF} = \frac{3}{2}$, so $EF = \frac{2 \times 12}{3}$

The triangles are similar and $EF = 8$ cm.

Example 2.2 In $\triangle ABC$, $\angle ABC = 90^\circ$ and $BD \perp AC$. Prove that $\triangle ABC \sim \triangle ADB$ and hence that $AB^2 = AD \times AC$.

SOLUTION

In $\triangle ABC$ and $\triangle ADB$: $\angle A$ is common, and $\angle ABC = \angle ADB = 90^\circ$.

So $\triangle ABC \sim \triangle ADB$ by AA.

$$\frac{AB}{AD} = \frac{AC}{AB}$$

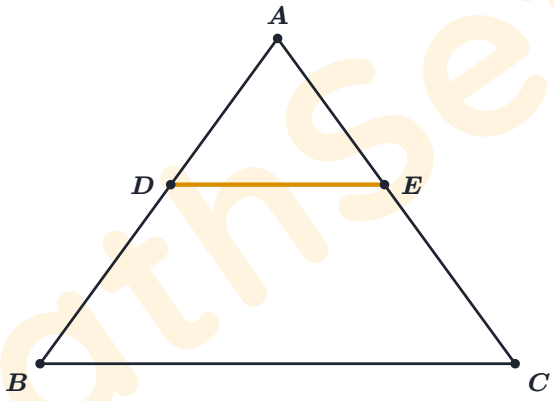
$$AB^2 = AD \times AC$$

3 Basic proportionality theorem

- **Theorem**: a line drawn parallel to one side of a triangle divides the other two sides proportionally.
- In $\triangle ABC$, if $DE \parallel BC$ with D on AB and E on AC:

$$\frac{AD}{DB} = \frac{AE}{EC} \qquad \frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

- **Converse**: a line that divides two sides of a triangle proportionally is parallel to the third side.



DE is parallel to BC, so AD : DB = AE : EC

Example 3.1 In $\triangle ABC$, $DE \parallel BC$ with $AD = 3$ cm, $DB = 5$ cm and $BC = 12$ cm. Find DE.

SOLUTION

$\triangle ADE \sim \triangle ABC$ (AA, since $DE \parallel BC$), so $\frac{DE}{BC} = \frac{AD}{AB}$

$AB = 3 + 5 = 8$ cm, so $\frac{DE}{12} = \frac{3}{8}$

$DE = 4.5$ **cm**

REMEMBER

$\frac{DE}{BC}$ equals $\frac{AD}{AB}$, the ratio with the **whole** side, not $\frac{AD}{DB}$.

4 Areas of similar triangles

- **Theorem**: the ratio of the areas of two similar triangles equals the ratio of the squares of their corresponding sides.

$$\frac{\text{area of } \triangle ABC}{\text{area of } \triangle PQR} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{CA^2}{RP^2}$$

- The same ratio holds for the squares of corresponding altitudes and medians.

Example 4.1 The areas of two similar triangles are 36 cm² and 81 cm². If a side of the smaller triangle is 4 cm, find the corresponding side of the larger.

SOLUTION

$\frac{36}{81} = \frac{4^2}{x^2}$, so $\frac{6}{9} = \frac{4}{x}$

$x = 6$ cm

Example 4.2 In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. Find the ratio of the area of $\triangle ADE$ to the area of trapezium $DBCE$.

SOLUTION

$\frac{AD}{AB} = \frac{2}{5}$, so $\frac{\text{area of } \triangle ADE}{\text{area of } \triangle ABC} = \frac{4}{25}$

If the area of $\triangle ADE$ is $4k$, the area of $\triangle ABC$ is $25k$ and the trapezium is $21k$.

The ratio is $4 : 21$.

5 Maps and models

- A map or a model is similar to the real object. The **scale factor** k is the ratio of a length on the map or model to the actual length.

$\text{length of model} = k \times \text{actual length}$ $\quad$ $\text{area of model} = k^2 \times \text{actual area}$ $\quad$ $\text{volume of model} = k^3 \times \text{actual volume}$

- A scale of $1 : 200$ means $k = \frac{1}{200}$.

Example 5.1 The scale of a model of a ship is $1 : 200$. (i) The model is 40 cm long; find the length of the ship. (ii) The deck of the model has area 250 cm²; find the area of the ship's deck in m². (iii) The volume of the model is 1 litre; find the volume of the ship in m³.

SOLUTION

(i) Actual length $= 40 \times 200 = 8000$ cm $= 80$ m

(ii) Actual area $= 250 \times 200^2 = 10{,}000{,}000$ cm² $= 1000$ m²

(iii) Actual volume $= 1 \times 200^3 = 8{,}000{,}000$ litres $= 8000$ m³

(i) 80 m (ii) 1000 m² (iii) 8000 m³

Example 5.2 On a map drawn to a scale of $1 : 25{,}000$, a rectangular plot measures 4 cm by 6 cm. Find its actual area in km².

SOLUTION

Actual length $= 4 \times 25000$ cm $= 1$ km and actual breadth $= 6 \times 25000$ cm $= 1.5$ km

Actual area $= 1.5$ km²

6 Common mistakes

- Matching the wrong pairs of sides because the vertices are not written in corresponding order.
- Using the ratio of sides for the ratio of areas. Areas need the **square** of the ratio; volumes need the **cube**.
- Forgetting to convert units: 1 $m^2 = 10{,}000$ cm^2, 1 km $= 100{,}000$ cm, 1 $m^3 = 1000$ litres.

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